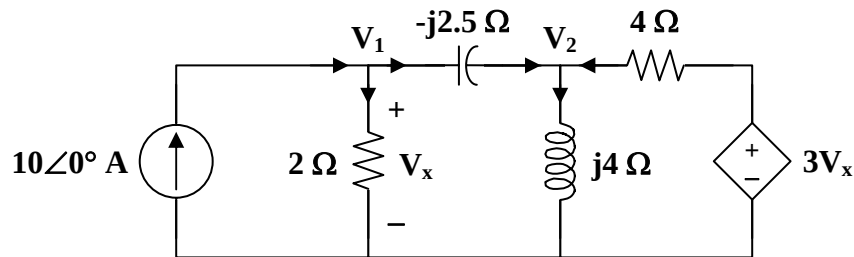


## CHAPTER 10

**P.P.10.1**      $10 \sin(2t) \longrightarrow 10 \angle 0^\circ, \quad \omega = 2$   
                    $2 \text{ H} \longrightarrow j\omega L = j4$   
                    $0.2 \text{ F} \longrightarrow \frac{1}{j\omega C} = -j2.5$

Hence, the circuit in the frequency domain is as shown below.



At node 1,  $10 = \frac{V_1}{2} + \frac{V_1 - V_2}{-j2.5}$   
 $100 = (5 + j4)V_1 - j4V_2$  (1)

At node 2,  $\frac{V_2}{j4} = \frac{V_1 - V_2}{-j2.5} + \frac{3V_x - V_2}{4}$      where  $V_x = V_1$   
 $-j2.5V_2 = j4(V_1 - V_2) + 2.5(3V_1 - V_2)$   
 $0 = -(7.5 + j4)V_1 + (2.5 + j1.5)V_2$  (2)

Put (1) and (2) in matrix form.

$$\begin{bmatrix} 5 + j4 & -j4 \\ -(7.5 + j4) & 2.5 + j1.5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \end{bmatrix}$$

where  $\Delta = (5 + j4)(2.5 + j1.5) - (-j4)(-(7.5 + j4)) = 22.5 - j12.5 = 25.74 \angle -29.05^\circ$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \frac{\begin{bmatrix} 2.5 + j1.5 & j4 \\ 7.5 + j4 & 5 + j4 \end{bmatrix}}{22.5 - j12.5} \begin{bmatrix} 100 \\ 0 \end{bmatrix}$$

$$V_1 = \frac{2.5 + j1.5}{22.5 - j12.5}(100) = \frac{2.915 \angle 30.96^\circ}{25.74 \angle -29.05^\circ}(100) = 11.32 \angle 60.01^\circ$$

$$V_2 = \frac{7.5 + j4}{22.5 - j12.5}(100) = \frac{8.5 \angle 28.07^\circ}{25.74 \angle -29.05^\circ}(100) = 33.02 \angle 57.12^\circ$$

In the time domain,

$$v_1(t) = \underline{11.32 \sin(2t + 60.01^\circ) \text{ V}}$$

$$v_2(t) = \underline{33.02 \sin(2t + 57.12^\circ) \text{ V}}$$

**P.P.10.2** The only non-reference node is a supernode.

$$\frac{15 - \mathbf{V}_1}{4} = \frac{\mathbf{V}_1}{j4} + \frac{\mathbf{V}_2}{-j} + \frac{\mathbf{V}_2}{2}$$

$$15 - \mathbf{V}_1 = -j\mathbf{V}_1 + j4\mathbf{V}_2 + 2\mathbf{V}_2$$

$$15 = (1 - j)\mathbf{V}_1 + (2 + j4)\mathbf{V}_2 \quad (1)$$

The supernode gives the constraint of

$$\mathbf{V}_1 = \mathbf{V}_2 + 20\angle 60^\circ \quad (2)$$

Substituting (2) into (1) gives

$$15 = (1 - j)(20\angle 60^\circ) + (3 + j3)\mathbf{V}_2$$

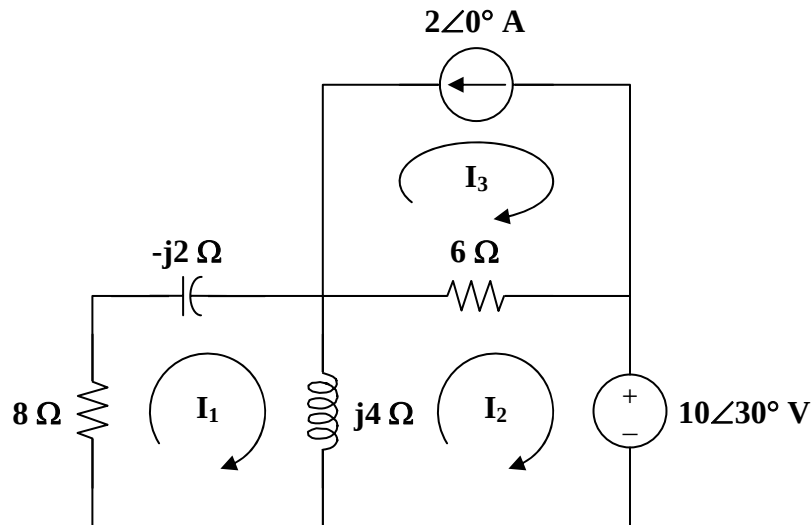
$$\mathbf{V}_2 = \frac{15 - (1 - j)(20\angle 60^\circ)}{3 + j3} = \frac{14.327\angle 210.72^\circ}{4.243\angle 45^\circ} = 3.376\angle 165.7^\circ$$

$$\mathbf{V}_1 = \mathbf{V}_2 + 20\angle 60^\circ = (-3.272 + j0.8327) + (10 + j17.32)$$

$$\mathbf{V}_1 = 6.728 + j18.154$$

Therefore,  $\mathbf{V}_1 = \underline{19.36\angle 69.67^\circ \text{ V}}$ ,  $\mathbf{V}_2 = \underline{3.376\angle 165.7^\circ \text{ V}}$

**P.P.10.3** Consider the circuit below.



For mesh 1,  $(8 - j2 + j4)\mathbf{I}_1 - j4\mathbf{I}_2 = 0$

$$(8 + j2)\mathbf{I}_1 = j4\mathbf{I}_2 \quad (1)$$

For mesh 2,  $(6 + j4)\mathbf{I}_2 - j4\mathbf{I}_1 - 6\mathbf{I}_3 + 10\angle 30^\circ = 0$

For mesh 3,  $\mathbf{I}_3 = -2$

Thus, the equation for mesh 2 becomes

$$(6 + j4)\mathbf{I}_2 - j4\mathbf{I}_1 = -12 - 10\angle 30^\circ \quad (2)$$

From (1),  $\mathbf{I}_2 = \frac{8 + j2}{j4}\mathbf{I}_1 = (0.5 - j2)\mathbf{I}_1 \quad (3)$

Substituting (3) into (2),

$$(6 + j4)(0.5 - j2)\mathbf{I}_1 - j4\mathbf{I}_1 = -12 - 10\angle 30^\circ$$

$$(11 - j14)\mathbf{I}_1 = -(20.66 + j5)$$

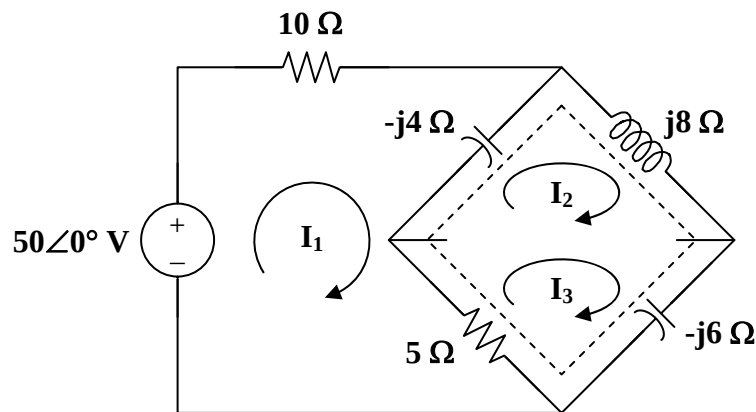
$$\mathbf{I}_1 = \frac{-(20.66 + j5)}{11 - j14}$$

Hence,  $\mathbf{I}_o = -\mathbf{I}_1 = \frac{20.66 + j5}{11 - j14} = \frac{21.256\angle 13.6^\circ}{17.8\angle -51.84^\circ}$

$$\mathbf{I}_o = \underline{\underline{1.194\angle 65.44^\circ \text{ A}}}$$

**P.P.10.4**

Mesches 2 and 3 form a supermesh as shown in the circuit below.



For mesh 1,  $-50 + (15 - j4)\mathbf{I}_1 - (-j4)\mathbf{I}_2 - 5\mathbf{I}_3 = 0$   
 $(15 - j4)\mathbf{I}_1 + j4\mathbf{I}_2 - 5\mathbf{I}_3 = 50 \quad (1)$

For the supermesh,  $(j8 - j4)\mathbf{I}_2 + (5 - j6)\mathbf{I}_3 - (5 - j4)\mathbf{I}_1 = 0 \quad (2)$

Also,  $\mathbf{I}_3 = \mathbf{I}_2 + 2 \quad (3)$

Eliminating  $\mathbf{I}_3$  from (1) and (2)

$$(15 - j4)\mathbf{I}_1 + (-5 + j4)\mathbf{I}_2 = 60 \quad (4)$$

$$(-5 + j4)\mathbf{I}_1 + (5 - j2)\mathbf{I}_2 = -10 + j12 \quad (5)$$

From (4) and (5),

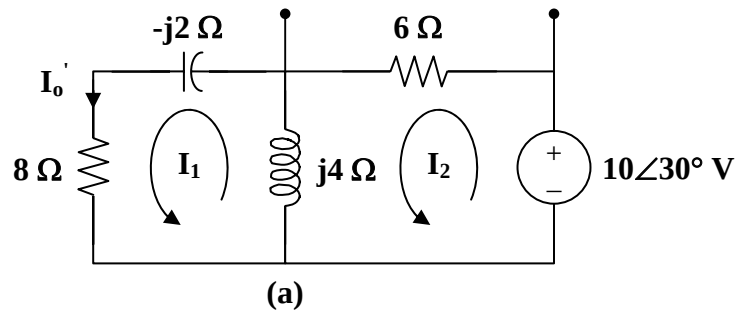
$$\begin{bmatrix} 15 - j4 & -5 + j4 \\ -5 + j4 & 5 - j2 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 60 \\ -10 + j12 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 15 - j4 & -5 + j4 \\ -5 + j4 & 5 - j2 \end{vmatrix} = 58 - j10 = 58.86 \angle -9.78^\circ$$

$$\Delta_1 = \begin{vmatrix} 60 & -5 + j4 \\ -10 + j12 & 5 - j2 \end{vmatrix} = 298 - j20 = 298.67 \angle -3.84^\circ$$

Thus,  $\mathbf{I}_o = \mathbf{I}_1 = \frac{\Delta_1}{\Delta} = \underline{\underline{5.074 \angle 5.94^\circ \text{ A}}}$

**P.P.10.5** Let  $\mathbf{I}_o = \mathbf{I}'_o + \mathbf{I}''_o$ , where  $\mathbf{I}'_o$  and  $\mathbf{I}''_o$  are due to the voltage source and current source respectively. For  $\mathbf{I}'_o$  consider the circuit in Fig. (a).



For mesh 1,  $(8 + j2)\mathbf{I}_1 - j4\mathbf{I}_2 = 0$   
 $\mathbf{I}_2 = (0.5 - j2)\mathbf{I}_1 \quad (1)$

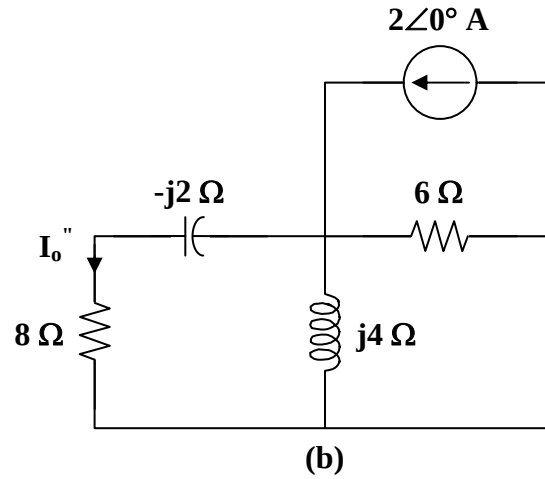
For mesh 2,  $(6 + j4)\mathbf{I}_2 - j4\mathbf{I}_1 - 10 \angle 30^\circ = 0 \quad (2)$

Substituting (1) into (2),

$$(6 + j4)(0.5 - j2)\mathbf{I}_1 - j4\mathbf{I}_1 = 10 \angle 30^\circ$$

$$\mathbf{I}'_o = \mathbf{I}_1 = \frac{10 \angle 30^\circ}{11 - j14} = 0.08 + j0.556$$

For  $\mathbf{I}_o''$  consider the circuit in Fig. (b).



$$\text{Let } \mathbf{Z}_1 = 8 - j2 \, \Omega, \quad \mathbf{Z}_2 = 6 \parallel j4 = \frac{j24}{6 + j4} = 1.846 + j2.769 \, \Omega$$

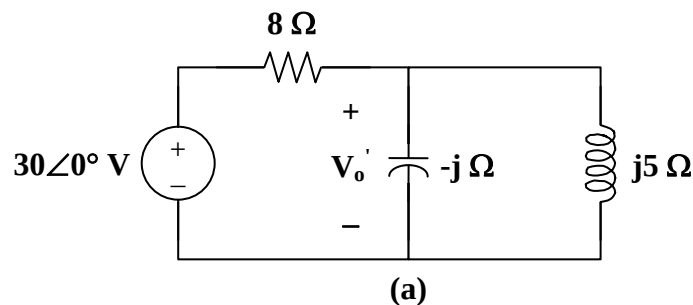
$$\mathbf{I}_o'' = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} (2) = \frac{(2)(1.846 + j2.769)}{9.846 + j0.77} = 0.4164 + j0.53$$

Therefore,  $\mathbf{I}_o = \mathbf{I}_o' + \mathbf{I}_o'' = 0.4961 + j1.086$   
 $\mathbf{I}_o = \underline{\underline{1.1939 \angle 65.45^\circ \text{ A}}}$

**P.P.10.6** Let  $v_o = v_o' + v_o''$ , where  $v_o'$  is due to the voltage source and  $v_o''$  is due to the current source. For  $v_o'$ , we remove the current source.

$$\begin{aligned} 30 \sin(5t) &\longrightarrow 30 \angle 0^\circ, \quad \omega = 5 \\ 0.2 \text{ F} &\longrightarrow \frac{1}{j\omega C} = \frac{1}{j(5)(0.2)} = -j \\ 1 \text{ H} &\longrightarrow j\omega L = j(5)(1) = j5 \end{aligned}$$

The circuit in the frequency domain is shown in Fig. (a).



Note that  $-j \parallel j5 = -j1.25$

By voltage division,

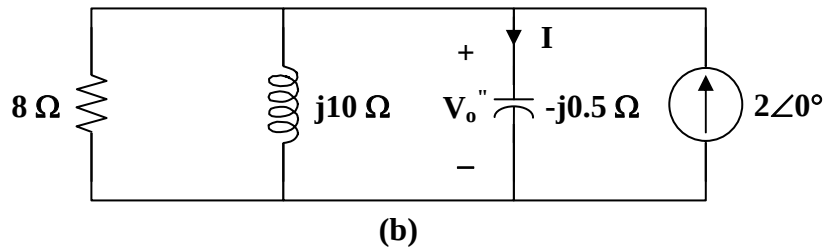
$$\mathbf{V}'_o = \frac{-j1.25}{8 - j1.25}(30) = 4.631 \angle -81.12^\circ$$

Thus,  $v'_o = 4.631 \sin(5t - 81.12^\circ)$

For  $v''_o$ , we remove the voltage source.

$$\begin{aligned} 2 \cos(10t) &\longrightarrow 2 \angle 0^\circ, \quad \omega = 10 \\ 0.2 \text{ F} &\longrightarrow \frac{1}{j\omega C} = \frac{1}{j(10)(0.2)} = -j0.5 \\ 1 \text{ H} &\longrightarrow j\omega L = j(10)(1) = j10 \end{aligned}$$

The corresponding circuit in the frequency domain is shown in Fig (b).



Let  $\mathbf{Z}_1 = -j0.5$ ,  $\mathbf{Z}_2 = 8 \parallel j10 = \frac{j80}{8 + j10} = 4.878 + j3.9$

By current division,

$$\mathbf{I} = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}(2)$$

$$\mathbf{V}''_o = \mathbf{I}(-j0.5) = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}(2)(-j0.5) = \frac{-j(4.877 + j3.9)}{4.878 + j3.4}$$

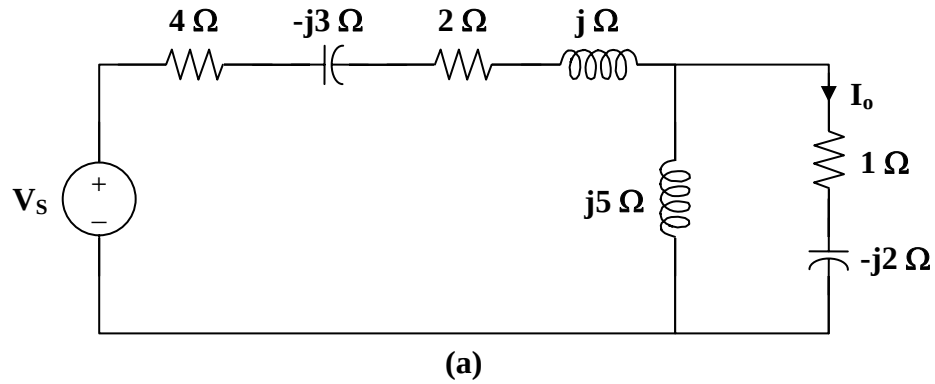
$$\mathbf{V}''_o = \frac{6.245 \angle -51.36^\circ}{5.94 \angle 34.88^\circ} = 1.051 \angle -86.24^\circ$$

Thus,  $v''_o = 1.051 \cos(10t - 86.24^\circ)$

Therefore,  $v_o = v'_o + v''_o$

$$v_o = \underline{\underline{4.631 \sin(5t - 81.12^\circ) + 1.051 \cos(10t - 86.24^\circ) \text{ V}}}$$

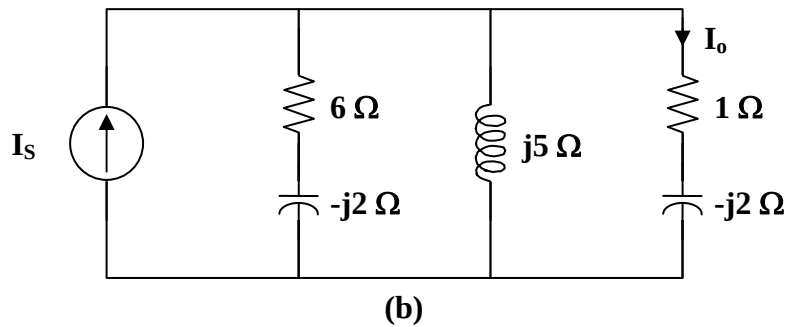
**P.P.10.7** If we transform the current source to a voltage source, we obtain the circuit shown in Fig. (a).



$$\mathbf{V}_s = \mathbf{I}_s \mathbf{Z}_s = (j4)(4 - j3) = 12 + j16$$

We transform the voltage source to a current source as shown in Fig. (b).

Let  $\mathbf{Z} = 4 - j3 + 2 + j = 6 - j2$ . Then,  $\mathbf{I}_s = \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{12 + j16}{6 - j2} = 1.5 + j3$ .



Note that  $\mathbf{Z} \parallel j5 = \frac{(6 - j2)(j5)}{6 + j3} = \frac{10}{3}(1 + j)$ .

By current division,

$$\begin{aligned} \mathbf{I}_o &= \frac{\frac{10}{3}(1 + j)}{\frac{10}{3}(1 + j) + (1 - j2)} (1.5 + j3) \\ \mathbf{I}_o &= \frac{-20 + j40}{13 + j4} = \frac{44.72 \angle 116.56^\circ}{13.602 \angle 17.1^\circ} \\ \mathbf{I}_o &= \underline{\underline{3.288 \angle 99.46^\circ \text{ A}}} \end{aligned}$$

**P.P.10.8**

When the voltage source is set equal to zero,

$$Z_{th} = 10 + (-j4) \parallel (6 + j2)$$

$$Z_{th} = 10 + \frac{(-j4)(6 + j2)}{6 - j2}$$

$$Z_{th} = 10 + 2.4 - j3.2$$

$$Z_{th} = \underline{12.4 - j3.2 \Omega}$$

By voltage division,

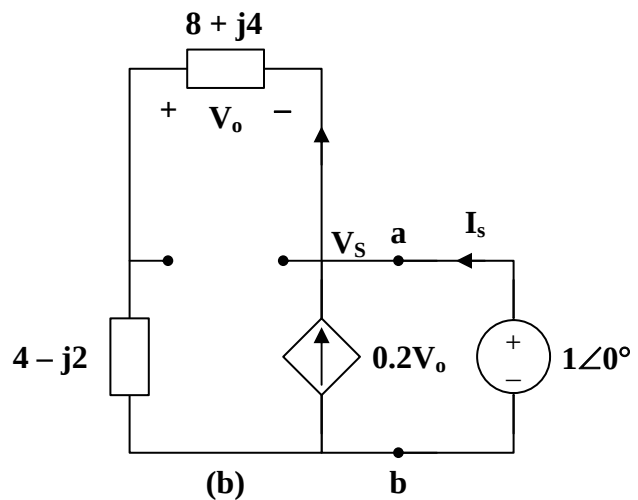
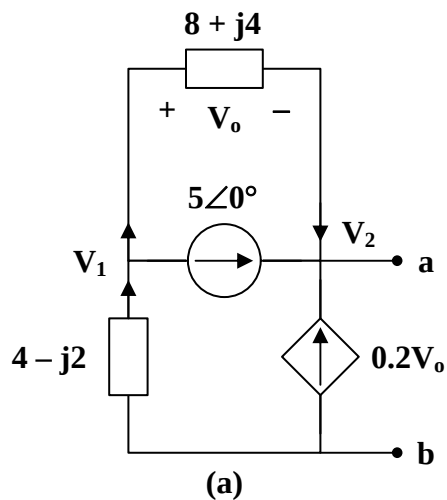
$$V_{th} = \frac{-j4}{6 + j2 - j4} (30 \angle 20^\circ) = \frac{(-j4)(30 \angle 20^\circ)}{6 - j2}$$

$$V_{th} = \frac{(4 \angle -90^\circ)(30 \angle 20^\circ)}{6.324 \angle -18.43^\circ}$$

$$V_{th} = \underline{18.97 \angle -51.57^\circ V}$$

**P.P.10.9**

To find  $V_{th}$ , consider the circuit in Fig. (a).



At node 1,

$$\frac{0 - V_1}{4 - j2} = 5 + \frac{V_1 - V_2}{8 + j4}$$

$$-(2 + j)V_1 = 50 + (1 - j0.5)(V_1 - V_2)$$

$$50 = (1 - j0.5)V_2 - (3 + j0.5)V_1 \quad (1)$$

At node 2,

$$5 + 0.2V_o + \frac{V_1 - V_2}{8 + j4} = 0,$$

where  $V_o = V_1 - V_2$ .

Hence, the equation for node 2 becomes

$$5 + 0.2(\mathbf{V}_1 - \mathbf{V}_2) + \frac{\mathbf{V}_1 - \mathbf{V}_2}{8 + j4} = 0$$

$$\mathbf{V}_1 = \mathbf{V}_2 - \frac{50}{3 + j0.5} \quad (2)$$

Substituting (2) into (1),

$$50 = (1 - j0.5)\mathbf{V}_2 - (3 + j0.5)\mathbf{V}_2 + (50) \frac{3 + j0.5}{3 - j0.5}$$

$$0 = -50 - (2 + j)\mathbf{V}_2 + \frac{50}{37}(35 + j12)$$

$$\mathbf{V}_2 = \frac{-2.702 + j16.22}{2 + j} = 7.35 \angle 72.9^\circ$$

$$\mathbf{V}_{th} = \mathbf{V}_2 = \underline{\underline{7.35 \angle 72.9^\circ \text{ V}}}$$

To find  $\mathbf{Z}_{th}$ , we remove the independent source and insert a 1-V voltage source between terminals a-b, as shown in Fig. (b).

At node a,  $\mathbf{I}_s = -0.2\mathbf{V}_o + \frac{\mathbf{V}_s}{8 + j4 + 4 - j2}$

But,  $\mathbf{V}_s = 1$  and  $-\mathbf{V}_o = \frac{8 + j4}{8 + j4 + 4 - j2} \mathbf{V}_s$

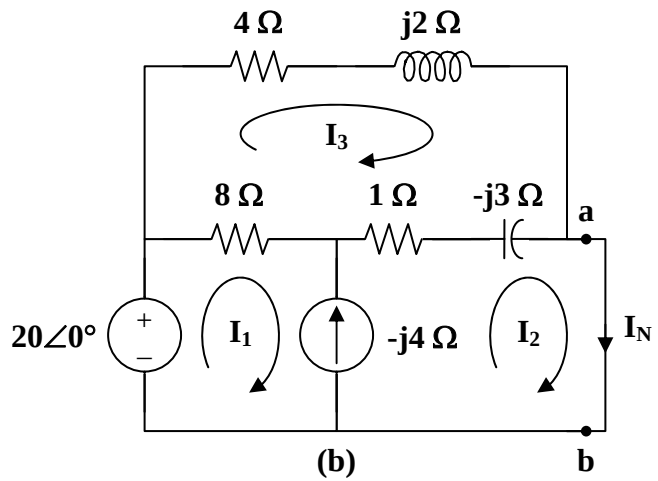
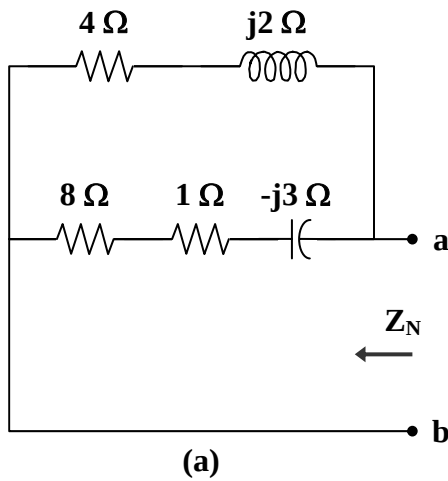
So,  $\mathbf{I}_s = (0.2) \frac{8 + j4}{12 + j2} + \frac{1}{12 + j2} = \frac{2.6 + j0.8}{12 + j2}$

and  $\mathbf{Z}_{th} = \frac{\mathbf{V}_s}{\mathbf{I}_s} = \frac{1}{\mathbf{I}_s} = \frac{12 + j2}{2.6 + j0.8} = \frac{12.166 \angle 9.46^\circ}{2.72 \angle 17.10^\circ}$

$$\mathbf{Z}_{th} = \underline{\underline{4.473 \angle -7.64^\circ \Omega}}$$

**P.P.10.10**

To find  $Z_N$ , consider the circuit in Fig. (a).



$$Z_N = (4 + j2) \parallel (9 - j3) = \frac{(4 + j2)(9 - j3)}{13 - j}$$

$$Z_N = \underline{\underline{3.176 + j0.706 \, \Omega}}$$

To find  $I_N$ , short-circuit terminals a-b as shown in Fig. (b). Notice that meshes 1 and 2 form a supermesh.

$$\text{For the supermesh,} \quad -20 + 8I_1 + (1 - j3)I_2 - (9 - j3)I_3 = 0 \quad (1)$$

$$\text{Also,} \quad I_1 = I_2 + j4 \quad (2)$$

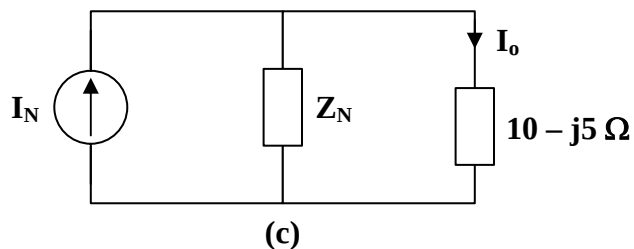
$$\text{For mesh 3,} \quad (13 - j)I_3 - 8I_1 - (1 - j3)I_2 = 0 \quad (3)$$

Solving for  $I_2$ , we obtain

$$I_N = I_2 = \frac{50 - j62}{9 - j3} = \frac{79.65 \angle -51.11^\circ}{9.487 \angle -18.43^\circ}$$

$$I_N = \underline{\underline{8.396 \angle -32.68^\circ \text{ A}}}$$

Using the Norton equivalent, we can find  $I_o$  as in Fig. (c).



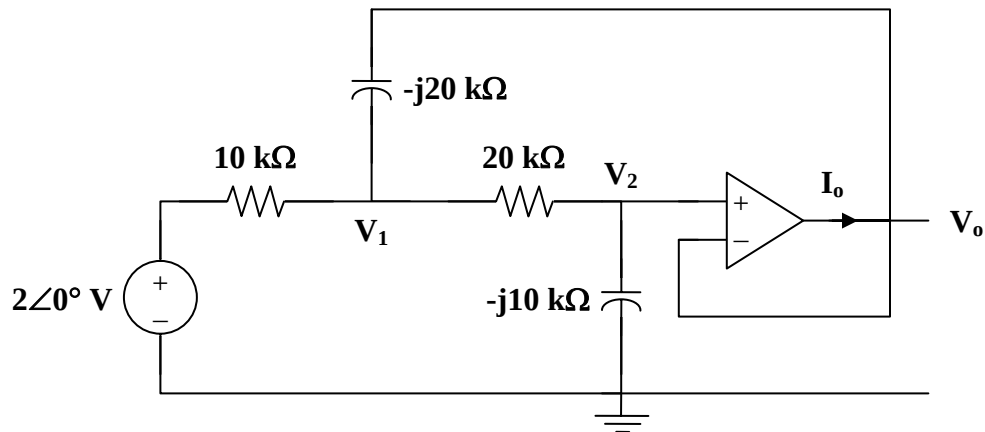
By current division,

$$\begin{aligned} \mathbf{I}_o &= \frac{\mathbf{Z}_N}{\mathbf{Z}_N + 10 - j5} \mathbf{I}_N = \frac{3.176 + j0.706}{13.176 - j4.294} (8.396 \angle -32.68^\circ) \\ \mathbf{I}_o &= \frac{(3.254 \angle 12.53^\circ)(8.396 \angle -32.68^\circ)}{13.858 \angle -18.05^\circ} \\ \mathbf{I}_o &= \underline{\underline{1.971 \angle -2.10^\circ \text{ A}}} \end{aligned}$$

**P.P.10.11**

$$\begin{aligned} 10 \text{ nF} &\longrightarrow \frac{1}{j\omega C_1} = \frac{1}{j(5 \times 10^3)(10 \times 10^{-9})} = -j20 \text{ k}\Omega \\ 20 \text{ nF} &\longrightarrow \frac{1}{j\omega C_2} = \frac{1}{j(5 \times 10^3)(20 \times 10^{-9})} = -j10 \text{ k}\Omega \end{aligned}$$

Consider the circuit in the frequency domain as shown below.



As a voltage follower,  $\mathbf{V}_2 = \mathbf{V}_o$

$$\begin{aligned} \text{At node 1, } \frac{2 - \mathbf{V}_1}{10} &= \frac{\mathbf{V}_1 - \mathbf{V}_o}{-j20} + \frac{\mathbf{V}_1 - \mathbf{V}_o}{20} \\ 4 &= (3 + j)\mathbf{V}_1 - (1 + j)\mathbf{V}_o \end{aligned} \quad (1)$$

$$\begin{aligned} \text{At node 2, } \frac{\mathbf{V}_1 - \mathbf{V}_o}{20} &= \frac{\mathbf{V}_o - 0}{-j10} \\ \mathbf{V}_1 &= (1 + j2)\mathbf{V}_o \end{aligned} \quad (2)$$

Substituting (2) into (1) gives

$$4 = j6\mathbf{V}_o \quad \text{or} \quad \mathbf{V}_o = \frac{2}{3} \angle -90^\circ$$

Hence,  $v_o(t) = 0.667 \cos(5000t - 90^\circ) \text{ V}$   
 $v_o(t) = \underline{\underline{0.667 \sin(5000t) \text{ V}}}$

Now,  $I_o = \frac{V_o - V_1}{-j20k}$

But from (2)  $V_o - V_1 = -j2V_o = \frac{-4}{3}$

$$I_o = \frac{-4/3}{-j20k} = -j66.66 \mu\text{A}$$

Hence,  $i_o(t) = 66.67 \cos(5000t - 90^\circ) \mu\text{A}$   
 $i_o(t) = \underline{\underline{66.67 \sin(5000t) \mu\text{A}}}$

**P.P.10.12** Let  $Z = R \parallel \frac{1}{j\omega C} = \frac{R}{1 + j\omega RC}$

$$\frac{V_s}{V_o} = \frac{R}{R + Z}$$

The loop gain is

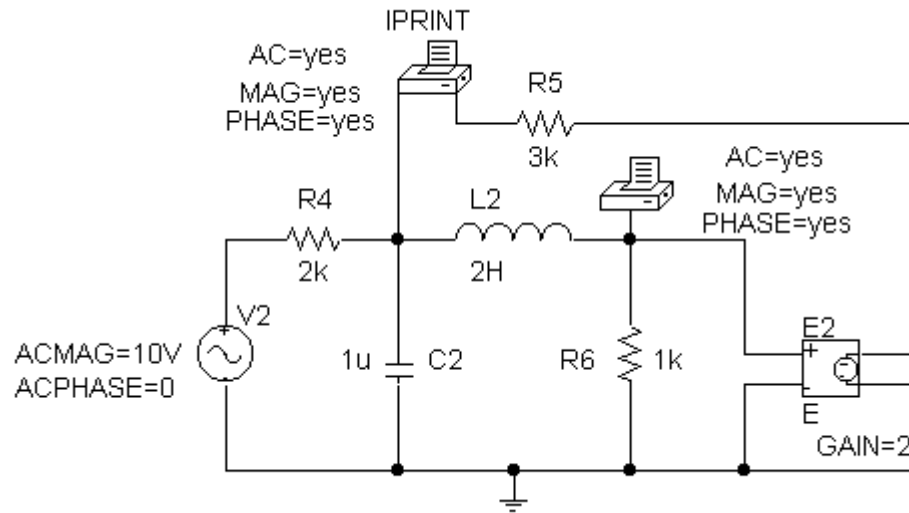
$$1/G = \frac{V_s}{V_o} = \frac{R}{R + Z} = \frac{R}{R + \frac{R}{1 + j\omega RC}} = \frac{1 + j\omega RC}{2 + j\omega RC}$$

where  $\omega RC = (1000)(10 \times 10^3)(1 \times 10^{-6}) = 10$

$$1/G = \frac{1 + j10}{2 + j10} = \frac{10.05 \angle 84.29^\circ}{10.2 \angle 78.69^\circ}$$

$$G = \underline{\underline{1.0147 \angle -5.6^\circ}}$$

**P.P.10.13** The schematic is shown below.



Since  $\omega = 2\pi f = 3000 \text{ rad/s} \longrightarrow f = 477.465 \text{ Hz}$ . Setup/Analysis/AC Sweep as Linear for 1 point starting and ending at a frequency of 477.465 Hz. When the schematic is saved and run, the output file includes

| Frequency | IM(V_PRINT1) | IP(V_PRINT1) |
|-----------|--------------|--------------|
| 4.775E+02 | 5.440E-04    | -5.512E+01   |
|           |              |              |
| Frequency | VM(\$N_0005) | VP(\$N_0005) |
| 4.775E+02 | 2.683E-01    | -1.546E+02   |

From the output file, we obtain

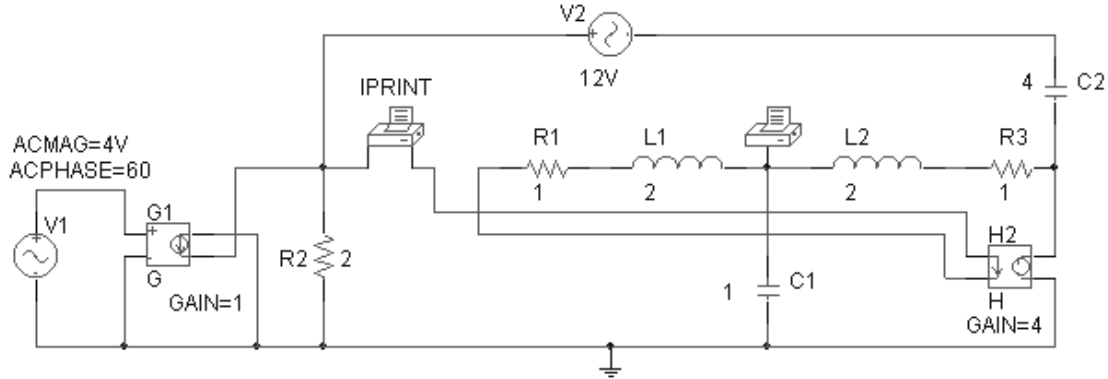
$$\mathbf{V_o = 0.2682 \angle -154.6^\circ \text{ V} \quad \text{and} \quad \mathbf{I_o = 0.544 \angle -55.12^\circ \text{ mA}}$$

Therefore,

$$\mathbf{v_o(t) = 0.2682 \cos(3000t - 154.6^\circ) \text{ V}}$$

$$\mathbf{i_o(t) = 0.544 \cos(3000t - 55.12^\circ) \text{ mA}}$$

**P.P.10.14** The schematic is shown below.



We select  $\omega = 1 \text{ rad/s}$  and  $f = 0.15915 \text{ Hz}$ . We use this to obtain the values of capacitances, where  $C = 1/\omega X_c$ , and inductances, where  $L = X_L/\omega$ . Note that IAC does not allow for an AC PHASE component; thus, we have used VAC in conjunction with G to create an AC current source with a magnitude and a phase. To obtain the desired output use Setup/Analysis/AC Sweep as Linear for 1 point starting and ending at a frequency of 0.15915 Hz. When the schematic is saved and run, the output file includes

| Frequency | IM(V_PRINT1) | IP(V_PRINT1) |
|-----------|--------------|--------------|
| 1.592E-01 | 2.584E+00    | 1.580E+02    |
| Frequency | VM(\$N_0004) | VP(\$N_0004) |
| 1.592E-01 | 9.842E+00    | 4.478E+01    |

From the output file, we obtain

$$\mathbf{V_x = 9.842 \angle 44.78^\circ \text{ V} \quad \text{and} \quad \mathbf{I_x = 2.584 \angle 158^\circ \text{ A}}$$

**P.P.10.15** 
$$C_{eq} = \left(1 + \frac{R_2}{R_1}\right)C = \left(1 + \frac{10 \times 10^6}{10 \times 10^3}\right)(10 \times 10^{-9}) = \mathbf{10 \mu F}$$

**P.P.10.16** If  $R = R_1 = R_2 = 2.5 \text{ k}\Omega$  and  $C = C_1 = C_2 = 1 \text{ nF}$

$$f_o = \frac{1}{2\pi RC} = \frac{1}{(2\pi)(2.5 \times 10^3)(1 \times 10^{-9})} = \mathbf{63.66 \text{ kHz}}$$